दिनांक .....
पृष्ठ .....

Week-1 module time complexity analysis

1) Square pattern

↳ Time complexity :- $O(n^2)$

The inner loop runs n times for each of the n rows.
total operations = $n \times n = n^2$

↳ Space complexity: O(1)
↳ We are not storing any value

2) Hollow Square, Right triangle, inverted triangle, pyramid, Diamond, Half-Diamond, Half-pi mond inverted, Number-Square, incrementing numb pattern, RTN, inverted Diamond number, Sandwich pattern, incrementing diamond pattern.

↳ All have same explanation like-above.

↳ nested loops have : $O(n^2)$ time complexity.

Week 2 Module

1) Linear Search

Complexity
- Best: O(1)
- Avg: O(n)
- Worst: O(n)
- Space: O(1)

Teacher's Signature.....

* Explanation.

↳ check elements one by one until target is found.

* optimal Solution.

↳ If array is unsorted, linear-search is already optimal

↳ If many searches are performed, use Hash-Table → O(1) Avg lookup

2) Binary Search.

Complexity

- Best: O(1)
- Avg: O(log n)
- Worst O(log n)
- Space:
  Iterative: O(1), Recursive: O(log n)

* Explanation

↳ Search space becomes half every iteration.

* Optimal Solution

↳ Binary search is already optimal for sorted arrays

3) Bubble-Sort

complexity

without optimisation

- best: $O(n^2)$
- Avg: $O(n^2)$
- Worst: $O(n^2)$

Teacher's Signature.....

optimal sol.
for real application use:

- Merge sort → o(n log n)
- quick sort → Avg o(n log n)
- Heap Sort → o(n log n)

4) Insertion sort:-

Complexity

- Best: O(n)
- Avg: O(n²)
- Worst: O(n²)

Space: O(1)

Explanation
Excellent for nearly sorted always

optimal Solution
- Nearly sorted → insertion sort
- Large random data → merge / quick sort.

5) Selection Sort
complexity
- Best: O(n²)
- Avg: O(n²)
- Worst: O(n²)

Space O(1)

Explanation

Teacher's Signature.....

finds minima every iteration.

* optimal sol
Not efficient for large datas*

1)	Merge sort	complexity Best: O(nlogn) Avg: O(nlogn) worst: O(nlogn) Space: O(n)	Explanation :- Divide array into halves and merge sorted halves. Optimal Sol merge sort itself is optimal
ii)	Merge Sort (max)	operation after sorting Median → O(1) Count > Median → O(n) Diff = max - min → O(1). Overall sorting dominates O(nlogn)	
iii)	quick Sort (Easy)	Best: O(nlogn) Avg: O(nlogn) Worst: O(n²) Space: O(logn) avg.	Explanation :- Partition around a pivot Optimal solution use randomised or median-of-three pivot to reduce the chance of the worst case.
iv)	quick sort (med	O(nlogn)	

Teacher's Signature.....

Heap sort	Best: O(n log n) Avg: o(n log n) Worst O(n log n) Space: O(1)	Explanation: Build Heap, then repeatedly remove the maximum element Optimal Sol: Heap sort is already optimal when constant extra space is req
Heap Sort(Med)	fastest - O(1) slowest - O(1) Avg - O(n) count below avg - O(n) percentage - 7(1) overall O(n log n)	

```
#include <bits/stdc++.h>
using namespace std;
int main() {
int N;
cin>> N;
for (int i= 1; i<=N; i++) {
cout << i << " ";
}
return 0;
}
```

i) Time complexity
O(N)

ii) Space complexity
O(1)

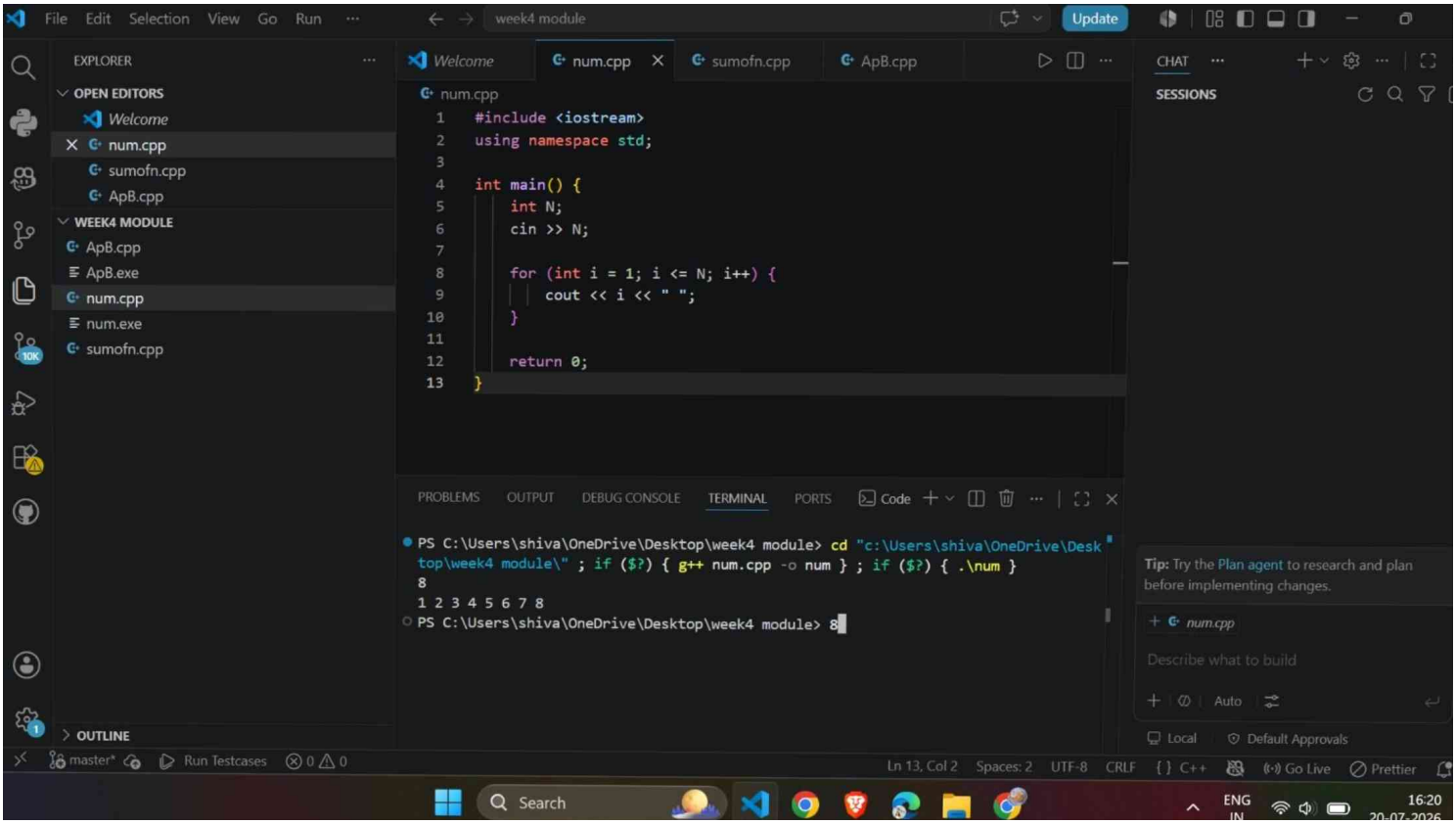
Explanation:

↳ The loop runs from 1 to N exactly once.
↳ Each iteration prints one number, so the total work is proportional to N.
↳ No extra data structures are used, so only constants extra memory is required.

Optimised approach

↳ The is already the optimal solution because every number from 1 to N must be printed.

Teacher's Signature.....


File Edit Selection View Go Run
week4 module
Update
EXPLORER
OPEN EDITORS
Welcome
num.cpp
sumofn.cpp
ApB.cpp
WEEK4 MODULE
ApB.cpp
ApB.exe
num.cpp
num.exe
sumofn.cpp
OUTLINE
Welcome
num.cpp
sumofn.cpp
ApB.cpp
num.cpp
#include <iostream>
using namespace std;
int main() {
int N;
cin >> N;
for (int i = 1; i <= N; i++) {
cout << i << " ";
}
return 0;
}
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Code
PS C:\Users\shiva\OneDrive\Desktop\week4 module> cd "c:\Users\shiva\OneDrive\Desktop\week4 module\" ; if ($?) { g++ num.cpp -o num } ; if ($?) { .\num }
8
1 2 3 4 5 6 7 8
PS C:\Users\shiva\OneDrive\Desktop\week4 module> 8
CHAT
SESSIONS
Tip: Try the Plan agent to research and plan before implementing changes.
num.cpp
Describe what to build
Auto
Local
Default Approvals
master*
Run Testcases
Ln 13, Col 2
Spaces: 2
UTF-8
CRLF
C++
Go Live
Prettier
Search
ENG
IN
16:20
20-07-2026

2) Sum of first N natural numbers using a loop.

```
#include <iostream>
using namespace std;

int main() {
long long N, sum = 0;
cin >> N;

for (long long i = 1; i <= N; i++) {
    sum += i;
}

cout << sum;
return 0;
}
```

Explanation

- The loop executes N times
- Each iteration performs one addition.
- only two variables (i and sum) are used.

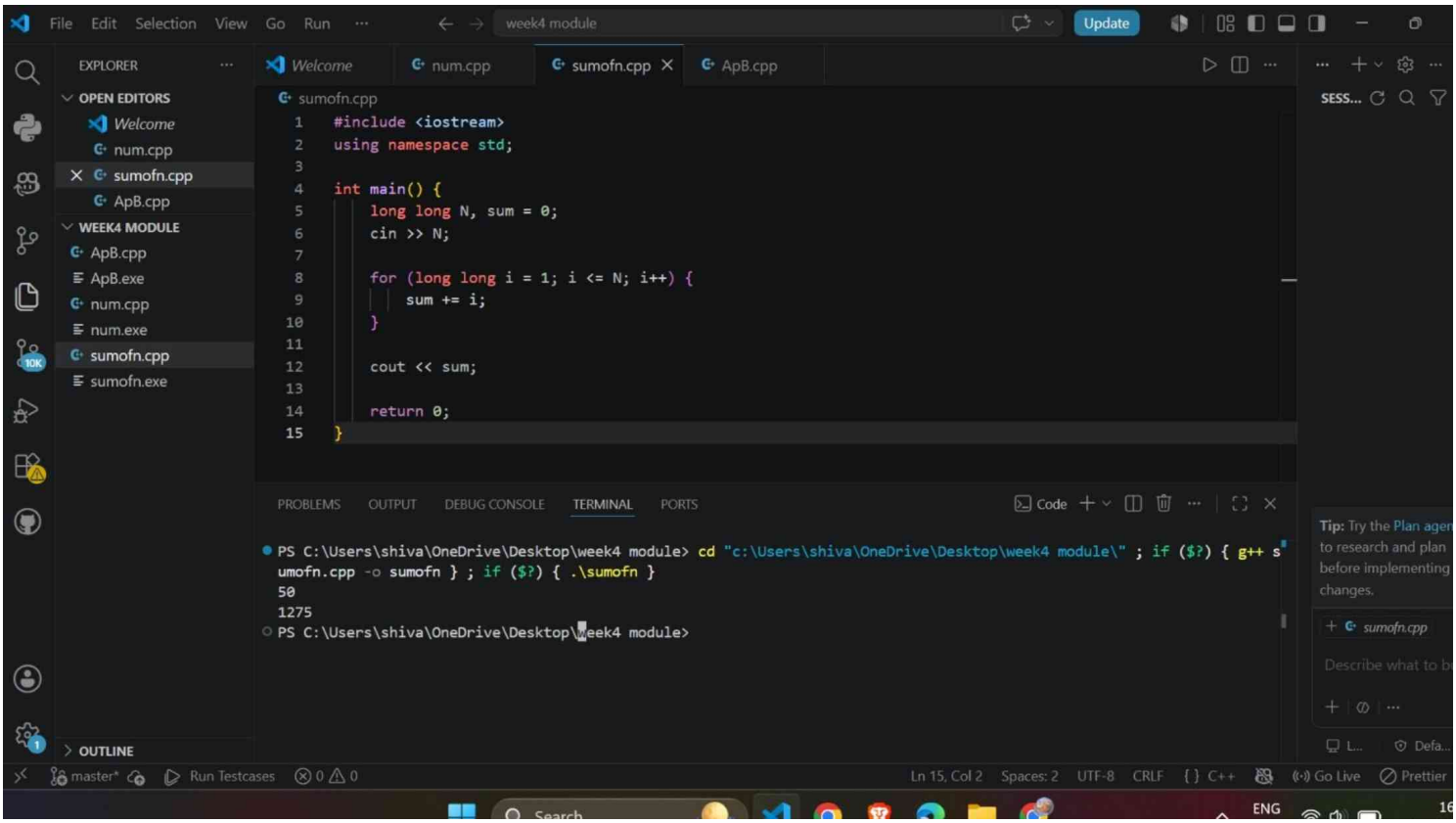
Optimised approach

we can use this formula

```
sum = N*(N+1)/2;
```

- Time complexity: O(1)
- Space complexity: O(1)

Teacher's Signature.....


File Edit Selection View Go Run
week4 module
Update
EXPLORER
OPEN EDITORS
Welcome
num.cpp
sumofn.cpp
ApB.cpp
WEEK4 MODULE
ApB.cpp
ApB.exe
num.cpp
num.exe
sumofn.cpp
sumofn.exe
OUTLINE
#include <iostream>
using namespace std;
int main() {
long long N, sum = 0;
cin >> N;
for (long long i = 1; i <= N; i++) {
sum += i;
}
cout << sum;
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Code
PS C:\Users\shiva\OneDrive\Desktop\week4 module> cd "c:\Users\shiva\OneDrive\Desktop\week4 module\" ; if ($?) { g++ s
umofn.cpp -o sumofn } ; if ($?) { .\sumofn }
50
1275
PS C:\Users\shiva\OneDrive\Desktop\week4 module>
Tip: Try the Plan ager
to research and plan
before implementing
changes.
sumofn.cpp
Describe what to b
master*
Run Testcases
Ln 15, Col 2
Spaces: 2
UTF-8
CRLF
C++
Go Live
Prettier
ENG

③ Calculate A^B using a loop.

```
#include <bits/stdc++.h>
using namespace std;

int main(){
long long A, B;
cin >> A >> B;
long long result = 1;
for(int i = 1; i <= B; i++){
    result *= A;
}
cout << result;
return 0;
}
```

Time complexity
O(B)
Space complexity
O(1)

Explanation

↳ The loop runs B time
↳ Each iteration performs one multiplication.
↳ Only a few variables are used, so the extra memory remains constant.

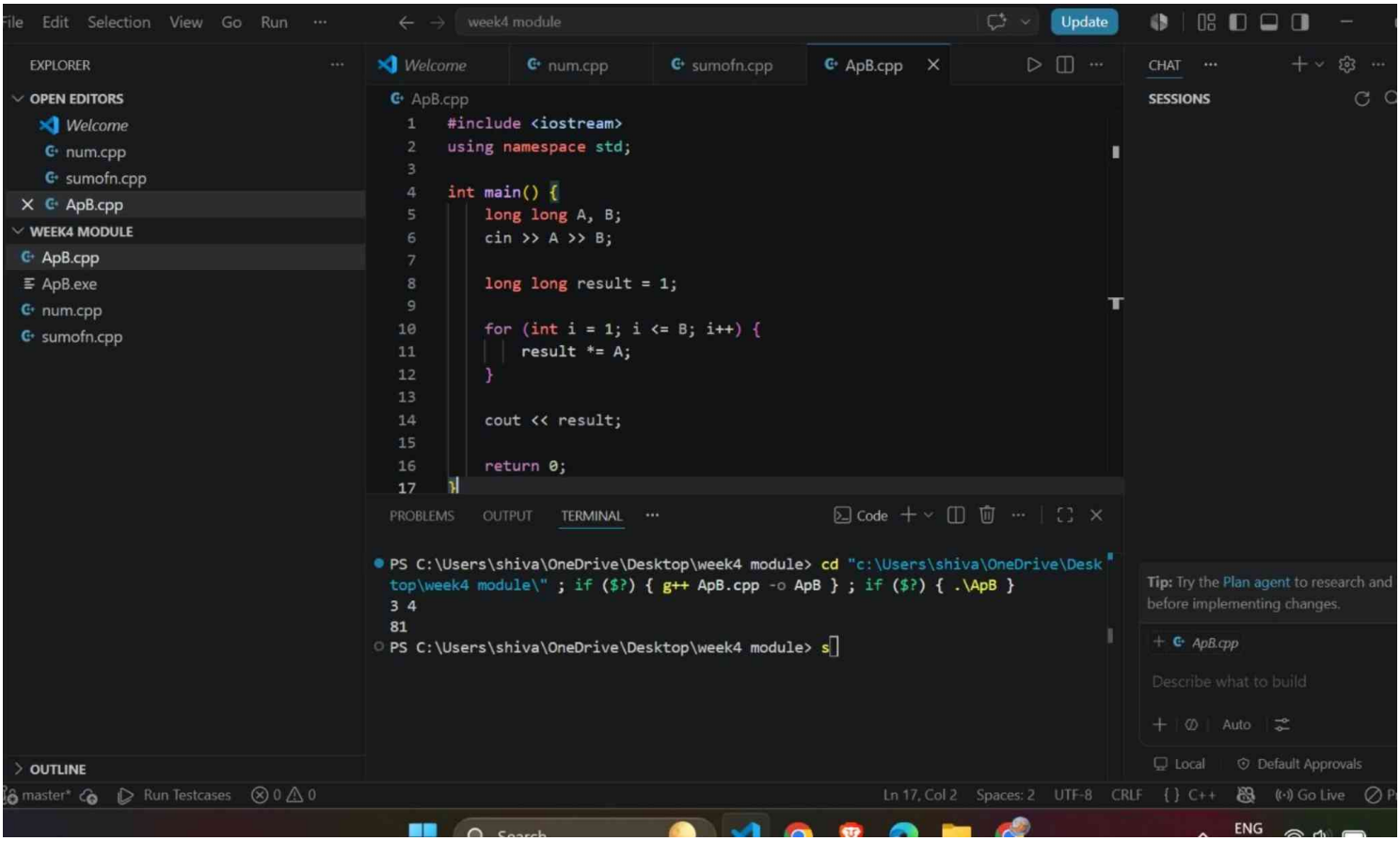
Teacher's Signature.....

दिनांक .....
पृष्ठ .....

optimised Approach.

use Binary Exponentation (fast powers)

- Time complexity: $O(\log B)$
- Space complexity: $O(1)$.


File Edit Selection View Go Run
week4 module
Update
EXPLORER
OPEN EDITORS
Welcome
num.cpp
sumofn.cpp
ApB.cpp
WEEK4 MODULE
ApB.cpp
ApB.exe
num.cpp
sumofn.cpp
OUTLINE
Welcome
num.cpp
sumofn.cpp
ApB.cpp
ApB.cpp
#include <iostream>
using namespace std;
int main() {
long long A, B;
cin >> A >> B;
long long result = 1;
for (int i = 1; i <= B; i++) {
result *= A;
}
cout << result;
return 0;
}
PROBLEMS
OUTPUT
TERMINAL
Code
PS C:\Users\shiva\OneDrive\Desktop\week4 module> cd "c:\Users\shiva\OneDrive\Desktop\week4 module\" ; if ($?) { g++ ApB.cpp -o ApB } ; if ($?) { .\ApB }
3 4
81
PS C:\Users\shiva\OneDrive\Desktop\week4 module> s
CHAT
SESSIONS
Tip: Try the Plan agent to research and before implementing changes.
ApB.cpp
Describe what to build
Auto
Local
Default Approvals
master*
Run Testcases
0 0
Ln 17, Col 2
Spaces: 2
UTF-8
CRLF
C++
Go Live
ENG